GRADUATION PROJECT REPORT

MediNova-AI

A Web-Based Healthcare Assistance System

GitHub Repository: https://github.com/Alaa-AHK/MediNova-AI

Team Members

Name	Email	Phone number
Alaa Abdelhakeem	alaaabdelhakeem47@gmail.com	01123199165
Salma zaghloul samak	salma.zaghloul56@gmail.com	01280700033
Afnan Magdy Eweis	Afnanmagdy20005@gmail.com	01141641440
Rokia Hassan Ahmed Mohamed	rokiahassan93@gmail.com	01152533383
Fady Youssif Esstemalk	gossef512@gmail.com	01228488639
Fares Waleed Badawi	fareselkhayat949@gmail.com	01223279568

Table of Contents

1. Project Planning & Management

1.1 Project Proposal

MediNova-AI is a comprehensive healthcare assistance platform designed to provide users with access to an extensive suite of AI-powered health-related services through a clean, modern web interface. Built using vanilla web technologies (HTML, CSS, JavaScript ES Modules), the project connects users to a wide variety of machine learning models covering diagnostic imaging, document OCR, conversational AI, and 3D visualization.

The project incorporates all of the following service modules:

- **Chest X-Ray Analysis:** Analyzes X-rays for pneumonia (ResNet50 + LoRA) and includes a 6-class hybrid classification model.
- **Pathology (Breast Cancer):** Provides histopathology classification (BreakHis via EfficientNetB5) as well as breast cancer image segmentation.
- **Skin Diseases:** Supports the identification of 7 common skin conditions based on dermoscopy images using a CNN (ResNet50).
- **Brain Tumor MRI Classification:** Analyzes brain MRI scans for tumor detection.
- **Heart Cancer Detection:** Explores heart CT segmentation and detection.
- **Eye Diseases Classification:** Detects various eye conditions from imaging data.
- **Kidney Disease Classification:** Trained on CT scans for kidney disease identification.
- **Prescription OCR:** Offers assistance in interpreting handwritten prescriptions utilizing TrOCR (VisionEncoderDecoder).
- **Medical Chatbot:** Connects to an external LLM API (e.g., via LangChain/Transformers) to provide conversational medical assistance.
- **3D Anatomical Visualization:** Integrates WebGL-based rendering of 3D medical or anatomical models (e.g., `.glb` assets) directly in the browser to assist with spatial understanding.

Each module is intended to serve as a supportive tool that complements the work of medical professionals, avoiding claims of replacing clinical diagnosis.

1.2 Objectives

1. To develop a web-based platform that allows users to access a massive suite of healthcare-related AI models from a single interface.
2. To successfully implement and integrate diverse ML workflows (CNNs, Transformers, LLMs, 3D rendering).
3. To design an intuitive, fast frontend utilizing vanilla HTML/JS without heavy frameworks.
4. To demonstrate the potential of technology-assisted healthcare support in an academic setting.

1.3 Scope

In Scope:

- A responsive, client-side web application with hash-based routing.
- Interfaces and APIs for extensive diagnostic models (Radiology, Pathology, Dermatology, Neurology, Cardiology, Nephrology, Ophthalmology).

- OCR functionality for digitized prescriptions.
- Interactive Medical Chatbot.
- In-browser 3D model visualization.

Out of Scope:

- Clinical validation or medical certification.
- Mobile application development (native app stores).
- Multi-language support or accessibility compliance beyond basic usability.

1.4 Project Timeline & 1.5 Task Assignment

The project team of six members worked collaboratively across dataset collection, algorithm selection (including CNNs, PEFT, and Transformers), system design, implementation, and testing phases.

1.6 Risk Management

- **Data Availability:** Mitigated by utilizing established datasets (HAM10000, BreakHis, CT Kidneys, etc.).
- **Integration Complexity:** Mitigated by decoupling the frontend (plain JS/HTML) from the backend APIs (FastAPI/Gradio/Streamlit), allowing independent model development.
- **Browser Compatibility (Camera/3D):** Camera API and 3D rendering require secure contexts (`localhost` or HTTPS), which is handled by proper local HTTP serving during demonstrations.

2. Literature Review

2.1 Overview of AI in Healthcare

AI-based tools have been explored for a wide range of medical tasks, including diagnostic imaging analysis, clinical decision support, and 3D surgical planning.

2.2 Diverse Radiology & Pathology Systems

Automated analysis of radiological and pathological scans has been studied extensively. MediNova-AI tackles this across multiple organs:

- **Chest X-Rays:** Binary pneumonia and 6-class hybrid classification.
- **Breast Cancer:** BreakHis histopathology and image segmentation.
- **Brain, Heart, & Kidney:** MRI and CT segmentation techniques for tumor/disease detection.
- **Eye Diseases:** Image classification models applied to ocular imagery.

2.3 Skin Disease Detection

Dermatology models trained on labeled skin lesion datasets (like HAM10000) can achieve high accuracy. MediNova-AI employs a CNN (ResNet50) to classify images into 7 classes (e.g., Melanoma, Basal Cell Carcinoma).

2.4 Prescription Assistance Systems (OCR)

Handwritten medical text recognition is notoriously difficult. MediNova-AI utilizes TrOCR (Transformer-based Optical Character Recognition), which relies on a Vision-Encoder-Decoder architecture to digitize handwritten prescriptions.

2.5 Medical Chatbots & LLMs

Large Language Models have opened new avenues for conversational clinical support. The platform integrates a chatbot module querying an external LLM API to provide conversational responses to medical inquiries.

2.6 3D Medical Visualization

The use of WebGL (e.g., Three.js) allows web applications to render complex 3D GLB/GLTF models, offering spatial insights into human anatomy or specific organ scans, enhancing the educational and diagnostic support value of the platform.

3. Requirements Gathering

3.1 Stakeholders

- **End Users:** Individuals interacting with the application to test the AI models and explore the 3D visualizer.
- **Project Team:** The six members responsible for developing the AI notebooks, training the models, and building the web application.
- **Instructors/Committee:** Reviewers of the graduation project.

3.2 User Stories

- **General:** As a user, I want to access a home page with clean navigation so I can select from the suite of AI services.
- **Medical Imaging (Radiology, Pathology, Brain, Heart, Eye, Kidney, Skin):** As a user, I want to upload an image or take a live photo using my device's camera, and receive a diagnostic prediction or probability distribution.
- **Prescription OCR:** As a user, I want to upload a prescription image and receive the transcribed text.
- **Chatbot:** As a user, I want to chat with an AI assistant for health-related queries.
- **3D Viewer:** As a user, I want to interact with a 3D anatomical model directly in my browser to explore its structure.

3.3 Functional Requirements

- **FR-1 UI & Navigation:** The system shall present a unified frontend with hash-based routing between all module pages.
- **FR-2 Media Upload & Camera:** The system shall support drag-and-drop file uploads and live camera capture (`getUserMedia`).
- **FR-3 Model Inference:** The system shall route media inputs to the respective backend APIs (Gradio, FastAPI) for all diagnostic models (X-Ray, Skin, Brain, etc.).
- **FR-4 Chat Interface:** The system shall provide a chat UI that maintains history and communicates with an external LLM.
- **FR-5 3D Rendering:** The system shall render `.glb` 3D assets interactively.

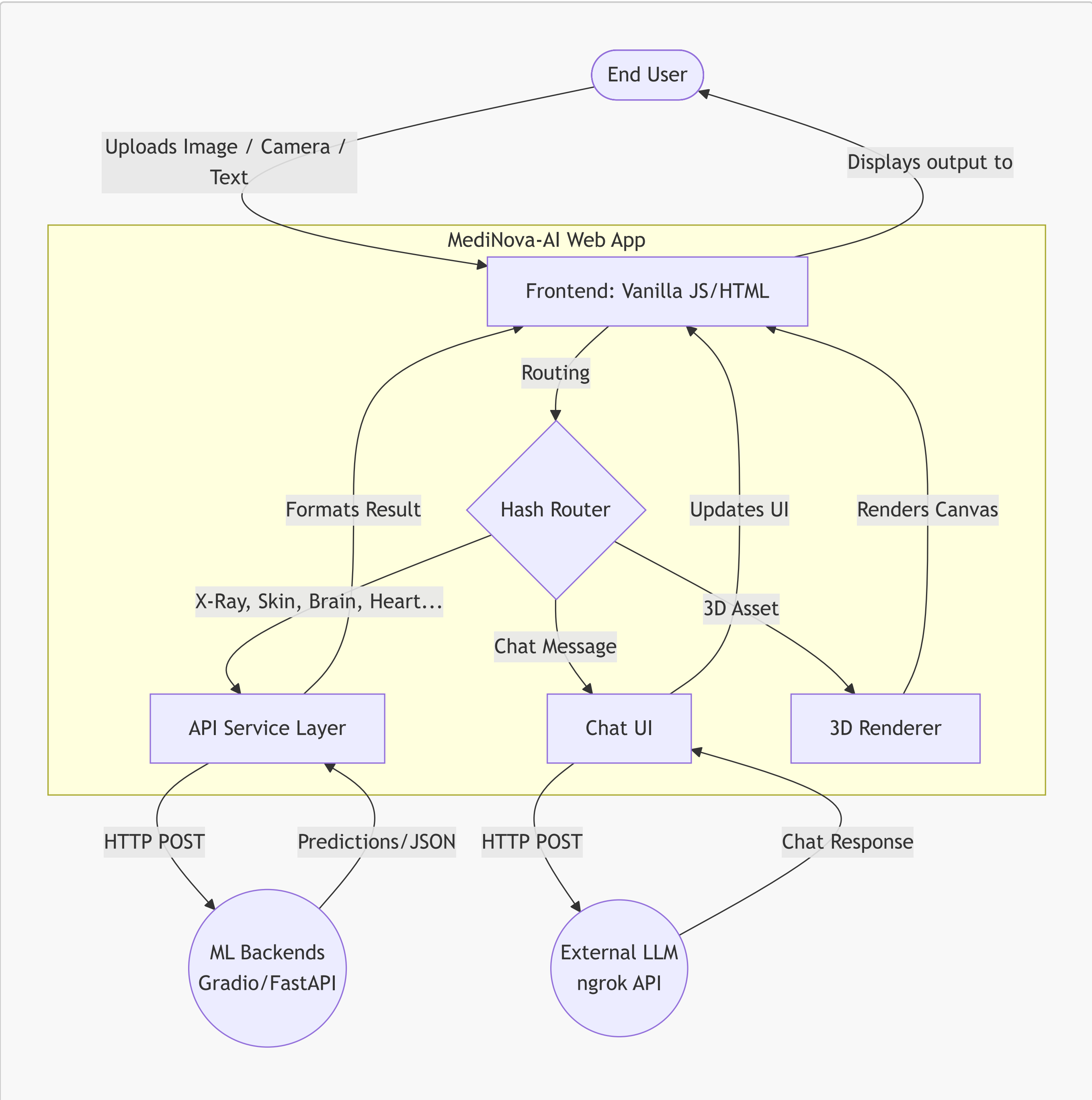
4. System Analysis & Design

4.1 System Architecture

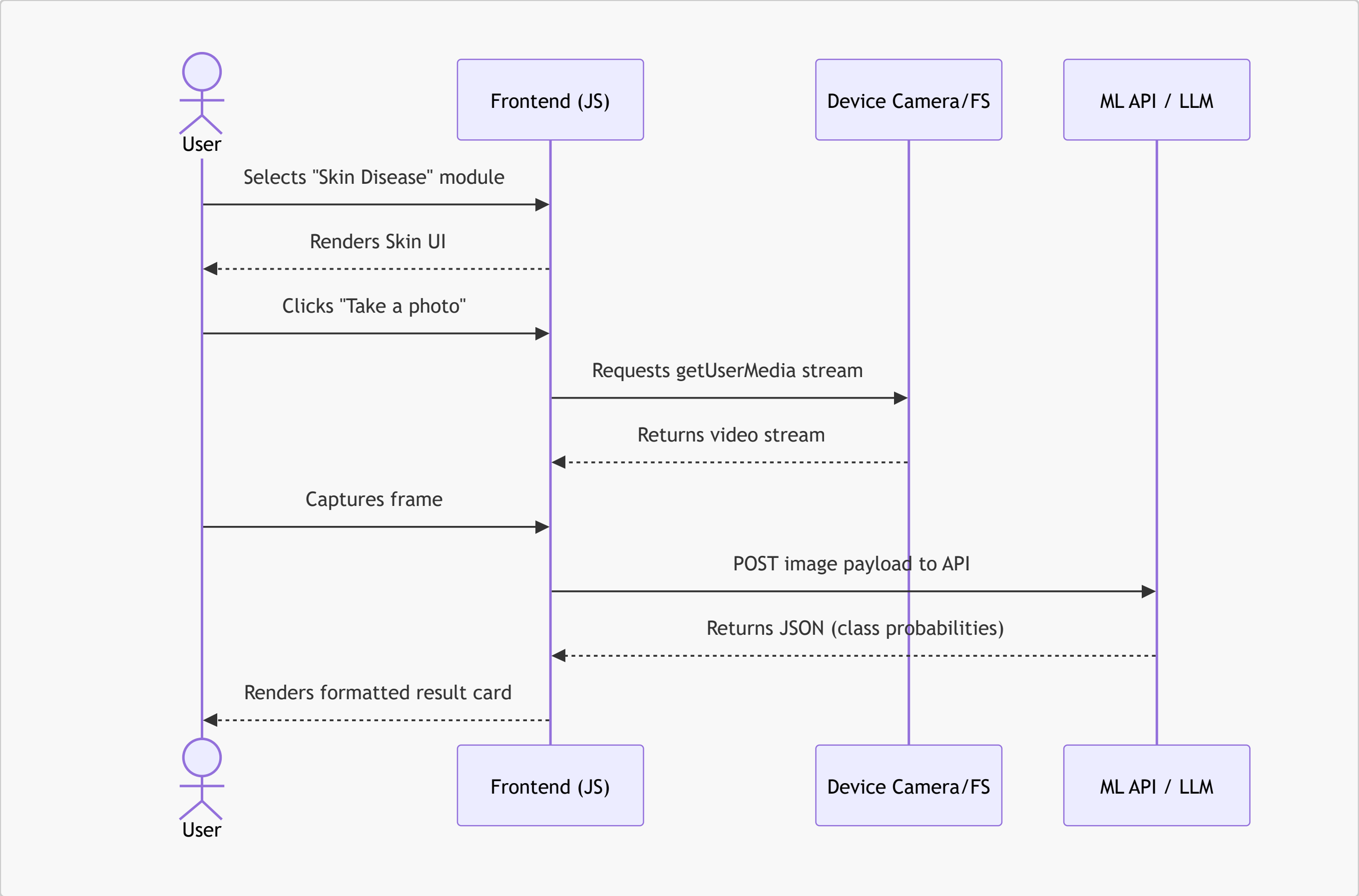
MediNova-AI utilizes a decoupled architecture to manage its massive suite of models:

- **Layer 1: Frontend UI (`Main_App`):** A lightweight, fast interface built with pure HTML5, CSS3, and JavaScript (ES Modules). Uses hash routing (`#/home`, `#/chatbot`) and generic API handlers.
- **Layer 2: AI Backends & Microservices:** The actual ML models reside in Python-based environments (Streamlit, Gradio Spaces, FastAPI, or ngrok tunnels). The frontend communicates with these services via REST APIs.

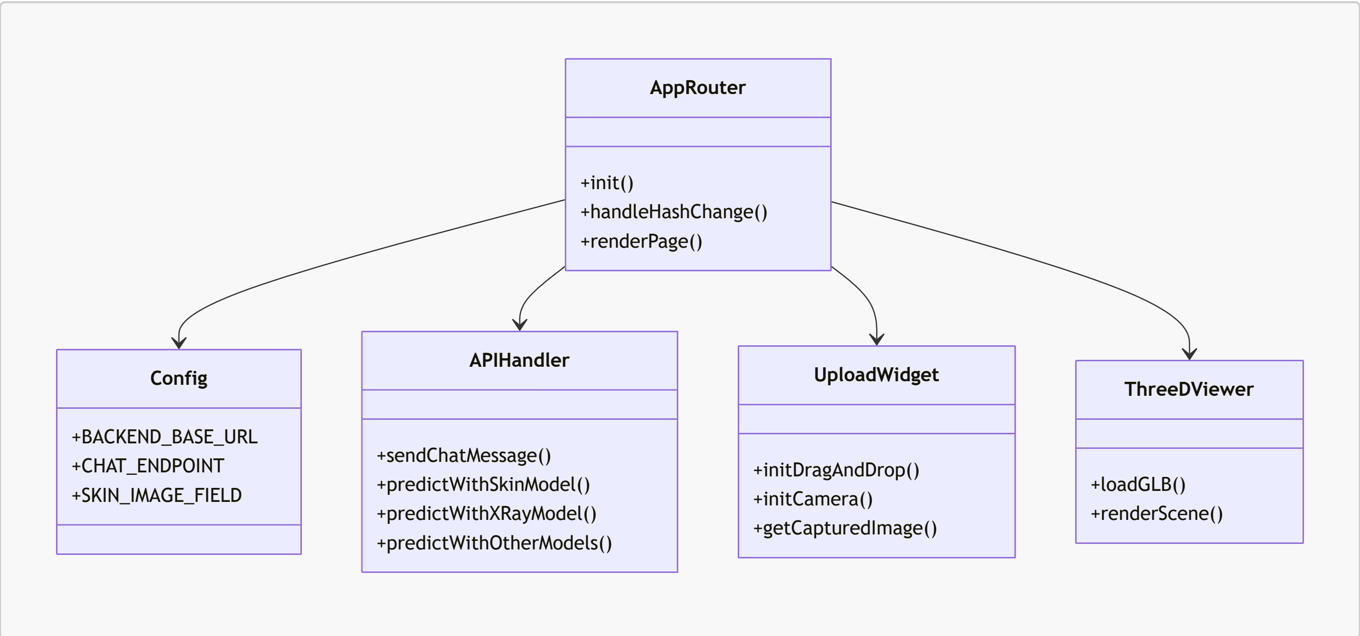
4.2 Data Flow Description (DFD)



4.3 Sequence Flow



4.4 Class / Component Diagram



4.5 Technology Stack

Layer	Technology	Purpose
Frontend UI	HTML5, CSS3, Vanilla JS (ES Modules)	Web interface, hash routing, camera capture
3D Rendering	WebGL / Three.js (Implied)	Rendering 3D GLB models in browser

Layer	Technology	Purpose
Backend Interfaces	Gradio, FastAPI, ngrok	Exposing ML models as REST APIs
Machine Learning	PyTorch, Keras/TensorFlow	Inference for all 9+ diagnostic models
NLP / Chat	Hugging Face Transformers, LLMs	TrOCR pipeline and Conversational AI

4.6 Deployment

Because the frontend uses ES Modules and requires a secure context for camera access, it must be served via a local web server rather than opening the HTML file directly. **Steps:**

1. Navigate to the `Main_App/medinova-ai` directory.
2. Start a local HTTP server: `python3 -m http.server 8000`.
3. Open `http://localhost:8000` in a browser.
4. (Optional) Run the local backend models and ensure `js/config.js` is updated with the correct endpoints (e.g., ngrok URLs for the Chatbot).